Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Butel, Janet S. Ph.D.		
Project Title:	Biology of Virus Infections: Radiation and Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	25
Comments:	Address updated 9/2008		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conner, Margaret (Baylor College of Medicine) Ling, Paul (Baylor College of Medicine) Zhang, Shaojie (Baylor College of Medicine)		
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Task Description:	Space flight has been found to affect immune responses, and alterations in a normal immune response often have a major impact on the host's ability to control infections. A question being explored is whether infectious diseases pose a medical risk to the success of long-duration space journeys. All humans are infected for life with latent and persistent viruses, and it is well-known that suppression of the immune system allows latent viruses to reactivate and multiply, which may cause disease in the person undergoing reactivation or in contacts to whom the virus is transmitted. The general hypothesis being addressed is that conditions of long-duration space flight, including radiation, stress, isolation, containment, and sleep deprivation will alter human immune responses, leading to reactivation of latent viruses, increased viral infections and viral disease, and possible development of malignancies. We are focusing on reactivation and shedding of human herpesvirus EBV and human polyomaviruses, agents known to establish persistent infections and to undergo reactivation and cause disease, including cancer, when the host immune system is compromised. Animal models are being used to study radiation effects on viral infections and host responses to those infections. The specific aims of the project are: (1) To determine the effects of space radiation and hind limb unloading (HLU) on host control of virus infections and virus-induced cancers; and (2) To characterize direct effects of radiation on viruses and virus-infected cells. Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. This model showed that gamma-irradiation leads to immunosuppression, delayed clearance of primary virus infection, and reactivated latent viral infections. Both single high-dose and multiple low doses of gamma-irradiation cause virus reactivation. HLU, that simulates aspects of weightlessness, was applied to the mouse model. HLU results in loss of control of virus infection in a tissue-specific manner. Combined effects of HLU and virus infection resulted in increased mortality. Plans for the coming year include experiments to further examine the effect of multiple, low-dose radiation exposures on host control of viral infections; expanded analysis of the effects of HLU and irradiation on control of polyoma virus infection, reactivation, and immune impairment; determination of cytokine and chemokine mRNA levels in tissues from HLU studies; measurements of virus reactivations in volunteers participating in the NASA-sponsored Bed Rest Study, a model for the effects of space flight conditions on human physiology, in collaboration with Dr. G. Sonnenfeld; and studies of radiation-induced signals that cause EBV reactivation from latency in cultured cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The mouse model we developed here involving polyoma virus infection and radiation will allow targeted studies of the effects of irradiation on host immune function, virus infection, and tumor development. Such data will define the risk of these combined factors to long-duration space flight and will allow tests of countermeasures. We have shown that radiation can cause reactivation of latent viral infections and depress host immune function. The evidence for viral reactivations highlights the need to understand synergistic effects of radiation and space flight conditions that may result in more deleterious effects on immune function and control of microbial infections and disease. The knowledge gained from studies of virus reactivations in these test models will be applicable to earth-bound individuals at risk of suffering similar virus reactivations and serious, sometimes life-threatening, consequences due to immunosuppression following organ transplantation or cancer chemotherapy and during pregnancy, old age, and AIDS. Studies of the role of EBV latent genes in human disease may lead to the development of novel antiviral drugs.
Task Progress:	Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. This model showed that gamma-irradiation leads to immunosuppression, delayed clearance of primary virus infection, and reactivated latent viral infections. Both single high-dose and multiple low doses of gamma-irradiation cause virus reactivation. Hind limb unloading (HLU), that simulates aspects of weightlessness, was applied to the mouse model. HLU results in loss of control of virus infection in a tissue-specific manner. Combined effects of HLU and virus infection resulted in increased mortality. These findings show the deleterious effect of multiple low-dose exposures to irradiation on host control of infections and the added adverse effect of HLU. This mouse model is proven useful to define the risk of combined factors that affect the immune system to long duration space flight.
Bibliography Type:	Description: (Last Updated: 04/25/2011)
Articles in Peer-reviewed Journals	Shearer WT, Zhang S, Reuben JM, Lee BN, Butel JS. "Effects of radiation and latent virus on immune responses in a space flight model." J Allergy Clin Immunol. 2005 Jun;115(6):1297-303. PMID: 15940150 , Jun-2005
Articles in Peer-reviewed Journals	Sonnenfeld G, Aviles H, Butel J S, Shearer W T, Niesel D, Pandya U, Allen C, Ochs H D, Blancher A, Abbal, M. "Bed rest and immunity." Acta Astronautica. In Press, 2006. Expected publication Feb 2007. , Apr-2006
Articles in Peer-reviewed Journals	Zhang S, McNees AL, Butel JS. "Quantification of vertical transmission of Murine polyoma virus by real-time quantitative PCR." J Gen Virol. 2005 Oct;86(Pt 10):2721-9. PMID: 16186225 , Oct-2005